DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/ Winter Examination – 2024

Course: F.Y. B.Tech **Branch : All Branches** **Semester : I**

Subject Code & Name: 24AF1CHEBS102 (Engineering Chemistry)

Max Marks: 60 **Date:** 08/02/2025 **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	----------- Indicator is used in Winkler's method of DO determination.				1	1
	a. Methanol	b. Starch	c. Cathol	d. Naphthol		
2	EBT makes ---------- dentate complex with Ca metal				1	1
	a. Bi	b. mono	c. tri	d. tetra		
3	Calorific value measured in -------				2	1
	a. ppm	b. ppb	c. mg/l	d. kcal/kg		
4	Boys Calorimeter is used to determine calorific value of ------- fuel.				2	1
	a. Gas	b. solid	c. wood	d. liquid		
5	Which of the following is an example of semi-solid lubricant?				2	1
	a. paint	b. alcohol	c. grease	d. diesel		
6	Cell constant is measured in ---------				3	1
	a. DO	b. MO	c. CO	d. none		
7	Specific conductance of KCl at 25 ^oC is ----------				3	1
	a.0.033	b. 0.0288	c. 0.273	d. 0.002765		
8	Color of Methyl Orange in alkali is -------				3	1
	a. green	b. yellow	c. red	d. orange		
9	Wavelength range of UV radiation is ---------				4	1
	a.700-800nm	b. 600-700nm	c. 600-400nm	d. 200-380nm		
10	Flame Photometer is based on ----------- of radiation.				4	1
	a. Substitution	b. Addition	c. Emission	d. refraction		

11	Which of the following is not an example of thermoplastic resin?				5	1
	a. Poly ethylene	b. Poly propylene	c. Poly styrene	d. Urea formaldehyde		
12	The Chemical formula of Gypsum is -----				5	1
	a. $MgSO_4$	b. $AgSO_4$	c. $CaSO_4.2H_2O$	d. FeSO4.2H2O		
Q. 2	**Solve the following.**					**12**
A)	Explain in detail Hot Lime-soda process of softening of water.				1	**6**
B)	Discuss aeration, sedimentation and disinfection process used in domestic water treatment.				1	**6**
Q.3	**Solve the following.**					**12**
A)	What is Calorific value? Explain in detail Bomb calorimeter.				2	**6**
B)	Describe any three Physical Properties of lubricants.				2	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	Explain Ostwald's theory of Acid-base indicator.				3	**6**
B)	Write a note on Conductometric titration with suitable examples.				3	**6**
C)	What is rechargeable battery? Explain in detail Lithium-ion battery.				3	**6**
Q.5	**Solve Any Two of the following.**					**12**
A)	Explain in detail Laws of absorption of UV-visible spectroscopy.				4	**6**
B)	What is Chromatography? Discuss the classification of Chromatography				4	**6**
C)	Discuss instrumentation, working and applications of Flame Photometry.				4	**6**
Q. 6	**Solve Any Two of the following.**					**12**
A)	Write a note on Portland Cement.				5	**6**
B)	Explain with suitable examples any two types of polymerization.				5	**6**
C)	Discuss the synthesis of Urea Formaldehyde resin, its properties and uses.				5	**6**
	***** End *****					